

STATE FORENSIC SCIENCE LABORATORY

JAIPUR, RAJASTHAN

FSL Report No.: FSL/JP/CIVIL/2025/0876 **Date:** 12 May 2025

Reference: FIR No. 214/2025, PS Civil Lines, Jaipur (dated 18.03.2025)

Samples received: 20.03.2025 under proper seal

To, The Investigating Officer, PS Civil Lines, Jaipur

Subject: Examination report of masonry mortar samples collected from the site of gallery collapse at Hemraj Memorial Stadium, Jaipur.

1. DESCRIPTION OF SAMPLES

- Sample A: Collapsed masonry mortar from North Gallery parapet (sealed packet marked "A")
- Sample B: Intact adjoining masonry mortar (sealed packet marked "B")
- Sample C: Control sample – not received

2. TESTS CONDUCTED

The samples were examined for compressive strength, consistency, and composition in accordance with the relevant Indian Standards for concrete and mortar.

3. OBSERVATIONS & RESULTS

- Visual: Sample A showed high porosity, poor bonding, and powdery residue on light abrasion.
- Compressive strength (28-day equivalent estimated): Sample A – 1.8 N/mm² (very low); Sample B – 4.2 N/mm².
- The mortar in Sample A does not meet the minimum requirements prescribed under **IS 1199:2018 (Methods of sampling and analysis of concrete)** for structural applications.
- Cement:sand ratio appeared leaner than 1:6.
- No conclusive evidence of deliberate adulteration, but workmanship and mix proportion appear deficient.

4. OPINION

The masonry mortar used in the collapsed portion (Sample A) is of sub-standard quality and does not conform to the specifications under IS 1199:2018. This deficiency is a probable contributing factor to the structural failure of the gallery parapet under normal load conditions present during the event.

5. This report is issued for the purpose of investigation only. The samples remain in the custody of FSL until further orders.

Sd/-

Dr. (Mrs.) Kavita Mehra
Assistant Director (Civil Engineering Division)
State FSL, Jaipur
Date: 12.05.2025

CRITICAL TESTING NOTE: This report deliberately applies the WRONG Indian Standard. IS 1199:2018 is for FRESH CONCRETE. The correct standard for masonry mortar is IS 2250:1981. Legal Luminaire is expected to flag this mismatch.
SYNTHETIC DOCUMENT — FOR TESTING LEGAL LUMINAIRE ONLY. No real incident or laboratory report.